
Learning a Parametric Embedding by Preserving Local Structure

Laurens van der Maaten

TiCC, Tilburg University

P.O. Box 90153, 5000 LE Tilburg, The Netherlands

lvdmaaten@gmail.com

Abstract

The paper presents a new unsupervised dimensionality reduction technique, called parametric t-SNE, that learns a parametric mapping between the high-dimensional data space and the low-dimensional latent space. Parametric t-SNE learns the parametric mapping in such a way that the local structure of the data is preserved as well as possible in the latent space. We evaluate the performance of parametric t-SNE in experiments on three datasets, in which we compare it to the performance of two other unsupervised parametric dimensionality reduction techniques. The results of experiments illustrate the strong performance of parametric t-SNE, in particular, in learning settings in which the dimensionality of the latent space is relatively low.

1 INTRODUCTION

The performance and efficiency of machine learning algorithms is often hampered by the high dimensionality of real-world datasets. Typically, the minimum number of parameters required to account for all properties of the data (i.e., the intrinsic dimensionality) is much smaller than the dimensionality of the data. Dimensionality reduction techniques try to exploit the relatively low intrinsic dimensionality of many real-world datasets. They embed the high-dimensional data in a latent space of lower dimensionality in such a way, that the structure of the data is retained as well as possible. Over the last decade, a large number of new non-parametric dimensionality reduction techniques have been proposed, such as Isomap (Tenenbaum et al., 2000), LLE (Roweis and Saul, 2000), and

MVU (Weinberger et al., 2004). The rationale behind these so-called manifold learners is that they attempt to retain the local structure of the data, that is, the small pairwise distances between the datapoints, in the latent space. The main limitation of the non-parametric manifold learners is that they do not provide a parametric mapping between the high-dimensional data space and the low-dimensional latent space, as a result of which the out-of-sample extension for these techniques is non-trivial. For spectral techniques such as the manifold learners listed above, the out-of-sample extension can be realized using the Nyström approximation (Bengio et al., 2004), but this leads to approximation errors and can become computationally expensive. As a result, the lack of a parametric mapping makes non-parametric dimensionality reduction techniques less suitable for use in, e.g., classification or regression tasks.

Despite the recent surge in non-parametric dimensionality reduction techniques, the development of new parametric dimensionality reduction techniques has been limited. Many parametric dimensionality reduction techniques, such as PCA and NCA (Goldberger et al., 2005), are hampered by their linear nature, which makes it difficult to successfully embed highly non-linear real-world data in the latent space. In contrast, autoencoders (Hinton and Salakhutdinov, 2006) can learn the non-linear mappings that are required for such embeddings, but they primarily focus on maximizing the variance of the data in the latent space, as a result of which autoencoders are less successful in retaining the local structure of the data in the latent space than manifold learners.

In this paper, we present a new unsupervised parametric dimensionality reduction technique that attempts to retain the local data structure in the latent space. The new technique, called parametric t-SNE, parametrizes the non-linear mapping between the data space and the latent space by means of a feed-forward neural network. Similar parametrizations have been proposed before, e.g., in NeuroScale (Lowe and Tipping, 1996) and back-constrained GPLVMs (Lawrence and Candela, 2006). The network is trained using a three-stage training procedure that is inspired by the training of autoencoders as described by Hinton and Salakhutdinov

Appearing in Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS) 2009, Clearwater Beach, Florida, USA. Volume 5 of JMLR: W&CP 5. Copyright 2009 by the authors.

通过保持局部结构学习参数嵌入

劳伦斯·范德马滕
蒂尔堡大学 TiCC, 邮政信箱
90153,5000 LE 蒂尔堡, 荷兰
lvdmaaten@gmail.com

抽象

论文提出了一种新的无监督降维技术, 称为参数 t-SNE, 它学习高维数据空间与低维潜空间之间的参数映射。参数 t-SNE 学习参数映射的方式, 使得数据的局部结构尽可能在潜在空间中保持完整。我们评估了参数化 t-SNE 在三个数据集实验中的表现, 并将其与另外两种无监督参数降维技术的表现进行比较。实验结果展示了参数化 t-SNE 的优异表现, 尤其是在潜空间维度相对较低的学习环境中。

1 介绍

机器学习算法的性能和效率常常受限于现实世界数据集的高维度。通常, 考虑数据所有属性 (即内在维度) 所需的最小参数数量远小于数据的维度。降维技术试图利用许多现实世界数据集相对较低的内在维度。他们将高维数据嵌入一个低维的潜在空间中, 以尽可能保留数据结构的方式。在过去十年中, 提出了大量新的非参数降维技术, 如 Isomap (Tenenbaum 等, 2000)、LLE (Roweis 和 Saul, 2000) 等。

MVU (Weinberger 等, 2004)。这些所谓流形学习器的背后原理是, 它们试图保留数据的局部结构, 即数据点之间在潜在空间中的小成对距离。非参数流形学习器的主要局限是它们无法在高维数据空间和低维潜空间之间提供参数映射, 因此这些技术的样本外扩展是非平凡的。对于上述流形学习器等谱技术, 样本外扩展可以通过 Nystrom 近似实现

(Bengio 等, 2004), 但这会导致近似误差, 且计算成本高昂。因此, 缺乏参数映射使得非参数降维技术不那么适合于诸如分类或回归任务。

尽管近年来非参数降维技术激增, 但新的参数降维技术的发展仍然有限。许多参数降维技术, 如 PCA 和 NCA

(Goldberger 等, 2005), 因其线性特性而受限, 这使得成功嵌入高度非线性现实世界数据在潜空间中变得困难。相比之下, 自编码器 (Hinton 和 Salakhutdinov, 2006) 可以学习此类嵌入所需的非线性映射, 但它们主要关注于最大化潜空间中数据的方差, 因此自编码器在保留潜在空间中数据局部结构的成功度不如流形学习器。

本文提出了一种新的无监督参数降维技术, 试图在潜在空间中保留局部数据结构。这种被称为参数化 t-SNE 的新技术, 通过前馈神经网络参数化数据空间与潜在空间之间的非线性映射。类似的参数化此前已有提出, 例如 NeuroScale

(Lowe 和 Tipping, 1996) 和背约束 GPLVM (Lawrence 和 Candela, 2006)。该网络采用三阶段训练过程, 灵感来源于 Hinton 和 Salakhutdinov 描述的自编码器训练

(2006). The three-stage training procedure aims to circumvent the problems of backpropagation procedures that are typically used to train neural networks.

The structure of the remainder of this paper is as follows. Section 2 introduces the new unsupervised parametric dimensionality reduction technique, called parametric t-SNE. The result of our experiments with parametric t-SNE on three datasets are presented in Section 3. The results are discussed in more detail in Section 4. Section 5 concludes the paper and presents directions for future research.

2 PARAMETRIC T-SNE

In parametric t-SNE, the parametric mapping $f : X \rightarrow Y$ from the data space X to the low-dimensional latent space Y is parametrized by means of a feed-forward neural network with weights W . We opt for the use of a (deep) neural network, because a neural network with sufficient hidden layers (with non-linear activation functions) is capable of parametrizing arbitrarily complex non-linear functions. The neural network is trained in such a way as to preserve the local structure of the data in the latent space. Herein, the cost function that is minimized in the training of the network is adapted from a recently introduced non-parametric dimensionality reduction technique, called t-SNE, that is good at visualizing the local structure of high-dimensional data (van der Maaten and Hinton, 2008).

The main problem of the training of deep neural networks is that the large number of weights (for typical problems about several millions) in the network cannot be learned successfully using backpropagation, as backpropagation tends to get stuck in poor local minima due to the complex interactions between the layers in the network. In order to circumvent this problem, we use a training procedure that is inspired by the training of autoencoders that is based on Restricted Boltzmann Machines (RBMs). The training procedure consists of three main stages: (1) a stack of RBMs is trained, (2) the stack of RBMs is used to construct a pre-trained neural network, and (3) the pre-trained network is finetuned using backpropagation as to minimize the cost function that attempts to retain the local structure of the data in the latent space. The training procedure of parametric t-SNE is illustrated in Figure 1. The pretraining (stage 1 and 2) and the finetuning (stage 3) of the parametric t-SNE network are discussed separately in 2.1 and 2.2.

2.1 PRETRAINING

The pretraining of a parametric t-SNE network consists of two stages. First, a stack of RBMs is trained. Second, the stack of RBMs is used to construct a pre-trained feed-forward neural network. Below, we first describe the training of an RBM (in 2.1.1). Subsequently, we turn to the procedure that constructs the pre-trained feed-forward neural network (in 2.1.2).

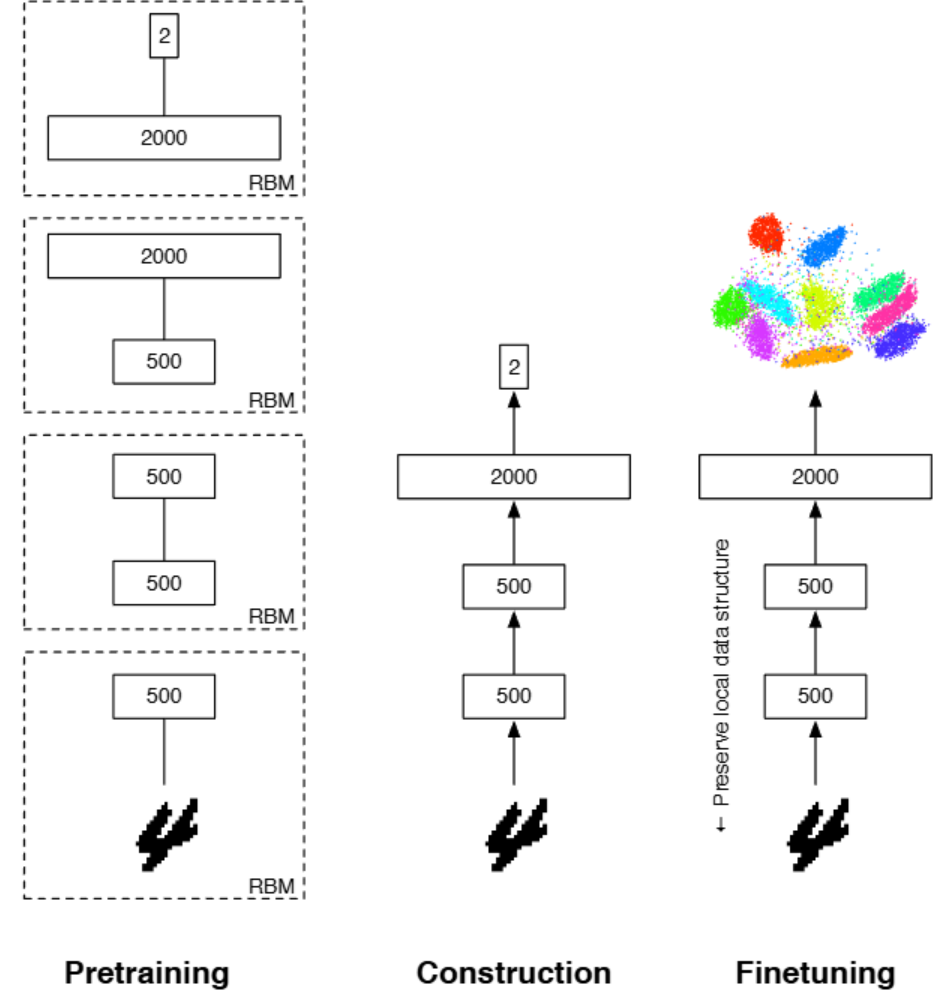


Figure 1: Overview of the three-stage training procedure of a parametric t-SNE network.

2.1.1 Restricted Boltzmann Machine

A Restricted Boltzmann Machine (RBM) is an undirected probabilistic graphical model, i.e., a Markov Random Field. The nodes of an RBM are usually Bernoulli distributed (Hinton, 2002), but if the mean field approximation is employed, the nodes may follow any exponential family distribution (Welling et al., 2004). The structure of an RBM is a fully connected bipartite graph, in which one group of nodes (the visual nodes $\mathbf{v}$) models the data, and the other group of nodes (the hidden nodes $\mathbf{h}$) models the latent structure of the data.

Since an RBM is a special case of a Markov Random Field, the joint distribution over all nodes is given by a Boltzmann distribution that is specified by the energy function $E(\mathbf{v}, \mathbf{h})$. The most common choice for the energy function is a linear function of the states of the visual and hidden nodes

$$E(\mathbf{v}, \mathbf{h}) = - \sum_{i,j} W_{ij} v_i h_j - \sum_i b_i v_i - \sum_j c_j h_j,$$

in which W_{ij} represents the weight of the connection between node v_i and h_j , b_i represents the bias on node v_i , and c_j represents the bias on node h_j . Noting that the states of the visual nodes are conditionally independent given the states of the hidden nodes and vice versa, it can easily be seen¹ that the linear energy function leads to conditional

¹Note that $p(\mathbf{v}|\mathbf{h}) = \frac{p(\mathbf{v}, \mathbf{h})}{\sum_{\mathbf{v}'} p(\mathbf{v}', \mathbf{h})}$, and that if we omit the biases, $p(\mathbf{v}, \mathbf{h}) \propto \exp(\mathbf{h}^T W \mathbf{v})$. Because v_i has a value of either 0 or 1, $p(v_i = 0|\mathbf{h}) = \frac{\exp(0)}{\exp(0) + \exp(\mathbf{h}^T W)} = \frac{1}{1 + \exp(\mathbf{h}^T W)}$.

(2006)。这一三阶段训练过程旨在规避通常用于训练神经网络的反向传播程序所带来的问题。

本文其余部分的结构如下。第二节介绍了新的无监督参数降维技术，称为参数 t-SNE。我们在三个数据集上参数化 t-SNE 实验的结果见第三节。结果将在第四节中有更详细的讨论。第五部分结束论文并提出了未来研究方向。

2 参数化 T-SNE

在参数化 t-SNE 中，参数映射 $f: X \rightarrow Y$ 从数据空间 X 到低维潜空间 Y ，通过权重为 W 的前馈神经网络参数化。我们选择使用（深度）神经网络，因为具有足够隐藏层（具有非线性激活函数）的神经网络能够参数化任意复杂的非线性函数。神经网络的训练方式是保持潜空间中数据的局部结构。在这里，网络训练中最小化的成本函数是从一种最近引入的非参数降维技术 t-SNE 改编而来，该技术在可视化高维数据的局部结构方面表现良好（van der Maaten 和 Hinton，2008）。

深度神经网络训练的主要问题在于，网络中权重数量庞大（典型问题约数百万个）无法通过反向传播成功学习，因为由于网络中层间复杂的相互作用，反向传播往往会卡在较差的局部极小值。为了绕过这个问题，我们采用了一种基于受限玻尔兹曼机（RBM）自编码器训练启发的训练程序。训练过程包括三个主要阶段：（1）训练一组 RBMs，（2）利用这些 RBMs 构建预训练神经网络，（3）通过反向传播对预训练神经网络进行微调，以最小化试图保留潜在空间中数据局部结构的成本函数。参数化 t-SNE 的训练过程如图 1 所示。参数化 t-SNE 网络的预训练（阶段 1 和 2）和微调（阶段 3）分别在 2.1 和 2.2 中讨论。

2.1 预培训

参数化 t-SNE 网络的预训练包含两个阶段。首先，训练一批 RBM。其次，利用 RBM 堆栈构建预训练的前馈神经网络。下面，我们首先描述 RBM 的训练方式（见 2.1.1）。随后，我们转向构建预训练前馈神经网络的过程（见 2.1.2）。

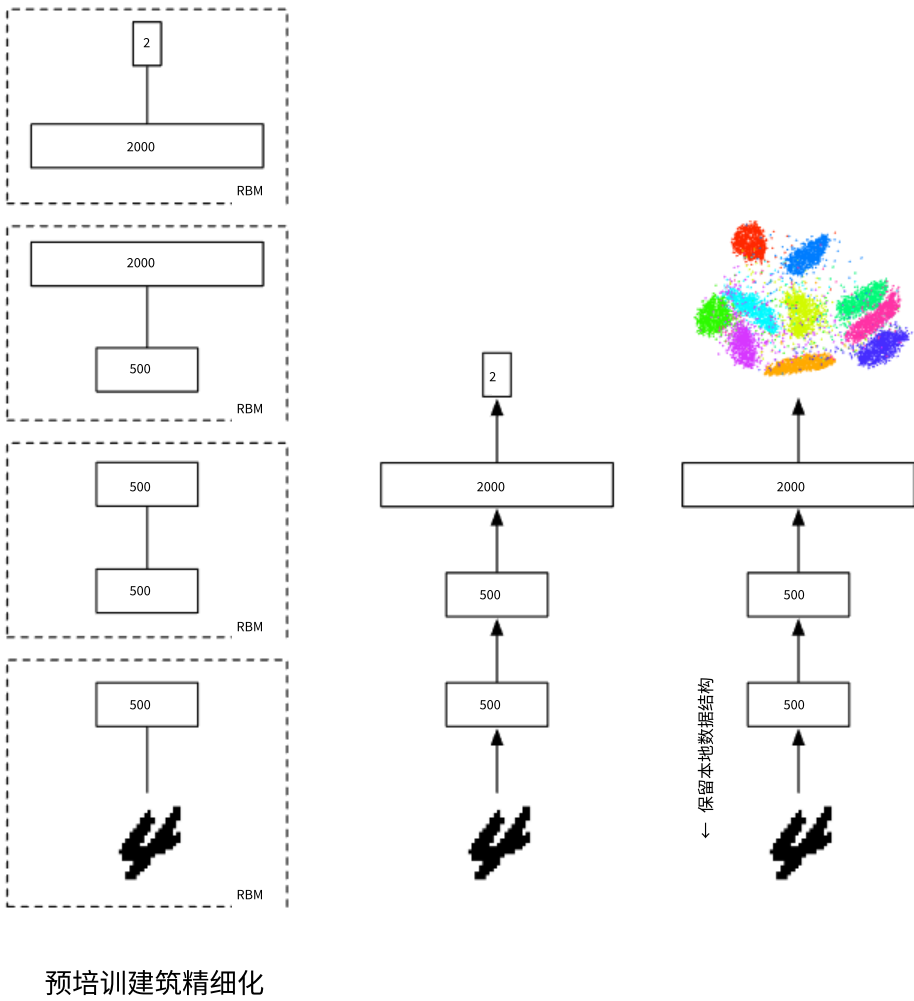


图 1：参数化 t-SNE 网络三阶段训练过程概述。

2.1.1 受限玻尔兹曼机

受限玻尔兹曼机（RBM）是一种无向概率图模型，即马尔可夫随机场。RBM 的节点通常具有伯努利分布（Hinton，2002），但如果采用均值场近似，节点可以遵循任何指数族分布（Welling 等，2004）。RBM 的结构是一个全连通的二分图，其中一组节点（视觉节点 v ）建模数据，另一组节点（隐藏节点 h ）建模数据的潜在结构。

由于 RBM 是马尔可夫随机场的特例，所有节点的联合分布由能量函数 $E(v, h)$ 指定的玻尔兹曼分布给出。能量函数最常见的选择是可见节点和隐藏节点状态的线性函数

$$E(v, h) = - \sum_{i, j} W_{ij} v_i h_j - \sum_i b_i v_i - \sum_j c_j h_j,$$

其中 W 代表节点 v 和 h 之间连接的权重， b 代表节点 V 上的偏置， c 代表节点 H 上的偏置。注意视觉节点的状态在条件独立于隐藏节点的状态，反之亦然，可以很容易看出线性能量函数导致条件函数

注意 $p(v|h) = P$ ，如果省略偏置， $p(v, h) \propto \exp(hWv)$ 。由于 v 的值要么是 0，要么是 1，

$$p(v=0|h) = \exp(0) / (\exp(0) + \exp(hWv))$$

probabilities $P(v_i = 1|\mathbf{h})$ and $P(h_j = 1|\mathbf{v})$ that are given by the sigmoid function of the input into a node

$$P(v_i = 1|\mathbf{h}) = \frac{1}{1 + \exp(-\sum_j W_{ij}h_j - b_i)}, \quad (1)$$

$$P(h_j = 1|\mathbf{v}) = \frac{1}{1 + \exp(-\sum_i W_{ij}v_i - c_j)}. \quad (2)$$

The weights W and the biases $\mathbf{b}$ and $\mathbf{c}$ of an RBM are learned in such a way that the marginal distribution over the visual nodes under the model, $P_{model}(\mathbf{v})$, is close to the observed data distribution $P_{data}(\mathbf{v})$. Specifically, the RBM is trained as to minimize the Kullback-Leibler divergence between the data distribution $P_{data}(\mathbf{v})$ and the model distribution $P_{model}(\mathbf{v})$, which is identical to maximizing the likelihood of the data under the model. The gradient of the Kullback-Leibler divergence with respect to the weights $W_{i,j}$ is given by

$$\frac{\delta KL(P_{data}||P_{model})}{\delta W_{ij}} = \mathbb{E}[v_i h_j]_{P_{data}} - \mathbb{E}[v_i h_j]_{P_{model}},$$

where $\mathbb{E}[\cdot]_{P_{model}}$ represents an expected value under the model distribution, and $\mathbb{E}[\cdot]_{P_{data}}$ represents an expected value under the data distribution.

Although the form of the gradient is fairly simple, it is impossible to compute the gradient, because the term $\mathbb{E}[v_i h_j]_{P_{model}}$ cannot be computed analytically. Sampling from the model distribution is also infeasible because this would require the Markov chain to be run infinitely long. In order to alleviate this problem, an alternative gradient has been proposed that minimizes a slightly different objective function that is called the *contrastive divergence* (Hinton, 2002). The contrastive divergence measures the tendency of the model distribution to *walk away* from the data distribution by $KL(P_{data}||P_{model}) - KL(P_1||P_{model})$, where $P_1(\mathbf{v})$ represents the distribution over the visual nodes as the RBM is allowed to run for one iteration (i.e., to perform one Gibbs sweep) when initialized according to the data distribution. The contrastive divergence can be minimized efficiently using standard gradient descent techniques, using an approximate gradient that is given by

$$\mathbb{E}[v_i h_j]_{P_{data}} - \mathbb{E}[v_i h_j]_{P_1}.$$

The term $\mathbb{E}[v_i h_j]_{P_1}$ is now estimated from samples that are obtained using Gibbs sampling (note that the required conditionals are given by Equation 1 and 2). The Markov chain of the sampler may be initialized by clamping a data vector onto the visual nodes, or by using the state of the Markov chain at the previous iteration.

2.1.2 Greedy Layer-Wise Training

The greedy layer-wise training procedure that is used to pretrain the parametric t-SNE network consists of three steps. First, the RBM that corresponds to the first layer

is trained on the input data (as described above). Second, the most likely values for the hidden nodes of the RBM are inferred for each datapoint. Third, these values are used as input data to train the RBM that corresponds to the second layer. This process is iterated for all layers in the network. The RBMs that correspond to the bottom layers of the neural network have Bernoulli-distributed hidden units, because this gives rise to a sigmoid activation function in the network. The RBM that corresponds to the top layer of the neural network uses Gaussian distributed hidden units, because this gives rise to a linear activation function in the network. The top layer of a neural network typically has a linear activation function to make the outputs of the network more stable.

The stack of trained RBMs is used to construct a pretrained parametric t-SNE network. Specifically, the undirected weights of the RBMs are untied and the biases on the visible units of the RBMs are dropped. As a result, the stack of RBMs is transformed into a pretrained feed-forward network. The resulting network forms a good initialization for the finetuning stage that aims to preserve the local structure of the data in the latent space (Larochelle et al., 2009). Preliminary experiments revealed that training parametric t-SNE networks without the pretraining stage leads to an inferior performance.

2.2 FINETUNING

In the finetuning stage, the weights of the pretrained neural network are finetuned in such a way that the network retains the local structure of the data in the latent space. This is done by converting the pairwise distances in both the data space and the latent space into probabilities that measure the similarity of two datapoints, and minimizing the Kullback-Leibler divergence between those probabilities (Hinton and Roweis, 2002, Min, 2005, van der Maaten and Hinton, 2008). Specifically, the pairwise distances in the data space are transformed into probabilities by centering an isotropic Gaussian over each datapoint i , computing the density of point j under this Gaussian, and renormalizing, yielding the conditional probabilities $p_{j|i}$

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)},$$

The variance of the Gaussian σ_i is set in such a way that the perplexity of each conditional distribution P_i is equal, and $p_{i|i}$ is set to zero. The perplexity is a free parameter that can be thought of as the number of effective neighbors. To form a single joint distribution, the conditional probabilities $p_{j|i}$ are symmetrized², i.e., we set $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$. The

²It is also possible to compute the joint probabilities p_{ij} directly by normalizing over all pairs of datapoints in Equation 2.2, however, such an approach gives inferior results under the presence of outliers.

由输入节点的 S 形函数给出的概率 $P(v=1|h)$ 和 $P(h=1|v)$

$$P(v=1|h) = \frac{1}{1 + \exp(-\sum_j W_{hj} - b)}, \quad (1)$$

$$P(h=1|v) = \frac{1}{1 + \exp(-\sum_i W_{vi} - c)}. \quad (2)$$

通过学习 RBM 中，视角节点的边际分布 $P(v)$ 接近观测数据分布 $P(v)$ ，从而学习权重 W 和偏差 c 。具体来说，RBM 训练以最小化数据分布 $P(v)$ 与模型分布 $P(v)$ 之间的 Kullback-Leibler 散度，这与最大化模型下数据的似然值相同。库尔巴克-莱布勒散度相对于权重 W_{is} 的梯度

$$\frac{\delta \text{KL}(P||P)}{\delta W} = E[vh] - E[vh],$$

其中 $E[\cdot]$ 表示模型分布下的期望值， $E[\cdot]$ 表示数据分布下的期望值。

虽然梯度的形式相当简单，但由于项 $E[vh]$ 无法解析计算，因此无法计算梯度。从模型分布抽样也不可行，因为这需要马尔可夫链无限长。为了缓解这个问题，提出了一种替代梯度，以最小化一个略有不同的目标函数，称为对比散度

(Hinton, 2002)。对比散度衡量模型分布偏离数据分布的趋势，即 $\text{KL}(P||P) - \text{KL}(P||P)$ ，其中 $P(v)$ 表示在 RBM 根据数据分布初始化时允许运行一次迭代（即执行一次 Gibbs 扫描）时，视觉节点上的分布。对比发散可以通过标准梯度下降技术高效地最小化，其近似梯度为

$$E[vh] - E[vh]_0.$$

项 $E[vh]$ 现在是通过吉布斯采样获得的样本估计的（注意所需的条件句由方程 1 和方程 2 给出）。采样器的马尔可夫链可以通过将数据向量夹在视觉节点上来初始化，或者使用上一次迭代的马尔可夫链状态来初始化。

2.1.2 贪婪的层级训练

用于预训练参数化 t-SNE 网络的逐层贪婪训练过程包含三个步骤。首先

对应第一层的 RBM 是基于输入数据进行训练的（如上所述）。其次，推断每个数据点对 RBM 隐藏节点的最可能值。第三，这些值作为输入数据，用于训练对应第二层的 RBM。该过程对网络中的所有层进行迭代。对应于神经网络底层的 RBMs 具有伯努利分布的隐藏单元，因为这导致网络中产生 S 形激活函数。对应于神经网络顶层的 RBM 使用高斯分布式隐藏单元，因为这在网络中产生线性激活函数。神经网络的顶层通常具有线性激活函数，以使网络输出更稳定。

训练好的 RBM 堆栈用于构建预训练的参数化 t-SNE 网络。具体来说，RBMs 的无向权重是解开的，RBM 可见单元的偏差也被消除。因此，RBM 堆栈被转化为预训练的前馈网络。由此产生的网络为微调阶段提供了良好的初始化，旨在保持数据在潜在空间中的局部结构 (Larochelle 等, 2009)。初步实验显示，未经过预训练阶段的参数化 t-SNE 网络训练会导致性能下降。

2.2 精细调化

在微调阶段，预训练神经网络的权重被微调，使网络保留潜在空间中数据的局部结构。这通过将数据空间和潜在空间中的两对距离转换为衡量两个数据点相似性的概率，并最小化这些概率之间的 Kullback-Leibler 散度来实现 (Hinton 和 Roweis, 2002; Min, 2005; van der Maaten 和 Hinton, 2008)。具体来说，数据空间中的两对距离通过将各向同性高斯分布置于每个数据点 i ，计算点 j 在该高斯分布下的密度，并重整化，得到条件概率 p ，从而将数据空间中的两两距离转换为概率

$$p = \frac{\exp(-\|x - x'\| / 2\sigma)}{\sum_{k=1}^K \exp(-\|x - x_k\| / 2\sigma)},$$

高斯分布 σ 的方差设置为每个条件分布的困惑度为 Π ， Π 为零。困惑度是一个自由参数，可以理解为有效邻居的数量。为了形成单一的联合分布，条件概率是 p -对称化的，即我们设 $p=2$ 。该

也可以通过对方程 2.2 中所有数据点对进行归一化来计算联合概率，但这种方法在存在离群值时结果较差。

resulting joint probabilities p_{ij} measure the similarity between datapoints i and j , as a result of which (assuming the variance of the Gaussians is relatively small) they capture the local structure of the data.

To measure the pairwise similarity of datapoints i and j in the latent space, a symmetric distribution is centered over each datapoint i in the latent space as well. Again, the density of all other points j under this distribution is measured, and the result is renormalized to obtain probabilities q_{ij} that represent the local structure of the data in the latent space. The weights of the parametric t-SNE network are now learned in such a way that the Kullback-Leibler divergence between the joint probability distributions P and Q is minimized, i.e., by minimizing

$$C = KL(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}. \quad (3)$$

The asymmetric nature of the Kullback-Leibler divergence leads the minimization to focus on modeling large p_{ij} 's by large q_{ij} 's. Hence, the objective function focuses on modeling similar datapoints close together in the latent space, as a result of which parametric t-SNE focuses on preserving the local structure of the data.

It may seem logical to use a Gaussian distribution to measure pairwise similarities q_{ij} in the latent space (as is done in, e.g., (Globerson et al., 2007, Hinton, 2002, Min, 2005, Iwata et al., 2007)), but this often leads to inferior results that are due to the *crowding problem* (van der Maaten and Hinton, 2008). The crowding problem is the result of the volume difference between high-dimensional and low-dimensional spaces that we explain in what follows.

Let us suppose that all small pairwise distances are retained perfectly in the low-dimensional latent space. Unless the original data lies in a subspace with an intrinsic dimensionality equal to or smaller than the dimensionality of the latent space, this implies that the larger pairwise distances cannot be modeled well in the latent space. In particular, the large pairwise distances have to be modeled as being larger. As a result, small attractive forces emerge between dissimilar datapoints in the latent space. The large number of such forces cause the crowding problem, as they 'crush' the data representation in the latent space together, which prevents the formation of separations between the natural classes in the data.

The crowding problem can be alleviated by using a heavy-tailed distribution to compute the pairwise similarities q_{ij} in the latent space. The use of a heavy-tailed distribution allows distant points to be modeled as being (too) far apart in the latent space, as a result of which the attractive forces that cause the crowding problem are eliminated. Because of its theoretical relation to the Gaussian distribution, we use a Student-t distribution as the heavy-tailed distribution to measure the pairwise similarities in the latent space. Denoting the mapping from the data space to the latent space that is defined by the feed-forward neural

network as $f : X \rightarrow Y$, this leads to the following definition of q_{ij}

$$q_{ij} = \frac{(1 + \|f(x_i|W) - f(x_j|W)\|^2/\alpha)^{-\frac{\alpha+1}{2}}}{\sum_{k \neq l} (1 + \|f(x_k|W) - f(x_l|W)\|^2/\alpha)^{-\frac{\alpha+1}{2}}}, \quad (4)$$

where α represents the number of degrees of freedom of the Student-t distribution. We discuss the appropriate setting of α later in this section.

The minimization of the cost function C (that uses the above definition of q_{ij}) can be performed using backpropagation, where the network is initialized using the procedure we described in 2.1. The gradient that is required for the finetuning is given by

$$\frac{\delta C}{\delta W} = \frac{\delta C}{\delta f(x_i|W)} \frac{\delta f(x_i|W)}{\delta W},$$

where $\frac{\delta f(x_i|W)}{\delta W}$ is computed using standard backpropagation, and $\frac{\delta C}{\delta f(x_i|W)}$ is given by

$$\frac{\delta C}{\delta f(x_i|W)} = \frac{2\alpha + 2}{\alpha} \sum_j (p_{ij} - q_{ij}) (f(x_i|W) - f(x_j|W)) (1 + \|f(x_i|W) - f(x_j|W)\|^2/\alpha)^{-\frac{\alpha+1}{2}}.$$

Because the number of p_{ij} 's and q_{ij} 's grows quadratically with the number of datapoints in the batch, the minimization of the cost function usually has to be performed using batches of a few thousand points (using larger batches is generally not possible because of memory constraints). The solution is updated after each gradient computation for a batch.

Now that we fully defined parametric t-SNE, we turn to the question of how the number of degrees of freedom α should be set. The Student-t distribution that is used in the latent space may contain a large portion of the probability mass under the distribution, because the volume of the latent space Y grows exponentially with its dimensionality. This leads to problems that may be addressed by setting the degrees of freedom α in such a way as to correct for the exponential growth of the volume of the latent space, because increasing the degrees of freedom α leads to a distribution with lighter tails. In fact, the parameter α determines to what extent the latent space is 'filled up': lower values of α lead to larger separations in the latent space between the natural clusters in the data, because they give rise to stronger repulsive forces between dissimilar datapoints in the latent space. In contrast, higher values of α lead to smaller separations between the natural clusters in the data, as a result of which more space is available in the latent space to appropriately model the local structure of the data. Below, we discuss three approaches to set the degrees of freedom of the Student-t distribution that is used to measure pairwise similarities in the latent space.

1) *Fixed value.* The first approach is to use a fixed setting

由此产生的联合概率 p 测量数据点 i 和 j 之间的相似性，因此（假设高斯方差相对较小）它们捕捉了数据的局部结构。为了测量潜空间中数据点 i 和 j 的两两相似度，也在潜空间中的每个数据点 i 上以对称分布为中心。同样，测量该分布下所有其他点 j 的密度，并对结果进行重整化，得到代表潜在空间中数据局部结构的概率 q 。参数化 t-SNE 网络的权重被学习得以最小化联合概率分布 P 和 Q 之间的 Kullback-Leibler 散度，即通过最小化

$$C = \text{KL}(P \parallel Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}. \quad (3)$$

Kullback-Leibler 散度的不对称性使得最小化重点是将大 p 建模为大 q 。因此，目标函数专注于在潜在空间中将相似数据点建模得很近，因此参数化 t-SNE 侧重于保持数据的局部结构。

使用高斯分布测量潜空间中的两两相似性看似合乎逻辑（如 Globerson 等，2007; Hinton, 2002; Min, 2005; Iwata 等，2007），但这通常导致因拥挤问题导致的结果较差

（van der Maaten 和 Hinton, 2008）。拥挤问题是高维空间与低维空间体积差异的结果，我们将在接下来的内容中解释。

假设所有小的两两距离在低维潜空间中都被完美保留。除非原始数据存在于内在维数等于或小于潜在空间维数的子空间中，否则这意味着较大的两两距离无法在潜空间中很好地建模。特别地，较大的两两距离必须被建模为更大的距离。因此，潜空间中不同数据点之间会产生微小的吸引力。大量此类力导致拥挤问题，因为它们在潜空间中“挤压”数据表示，阻碍了数据中自然类别之间的分离形成。

拥挤问题可以通过使用重尾分布计算潜空间中的两两相似度 q 来缓解。使用重尾分布可以将远点建模为潜空间中距离过远，从而消除导致拥挤问题的吸引力。由于其与高斯分布的理论关系，我们使用学生-t 分布作为重尾分布来衡量潜空间中的两两相似性

将前馈神经网络定义的数据空间到潜在空间的映射记为 $f: X \rightarrow Y$ ，这得出 q 的如下定义

$$q = \frac{\left(\frac{1}{1 + \|f(x|W) - f(x|W)\|/\alpha} \right)}{\sum_{k \neq l} \left(\frac{1}{1 + \|f(x|W) - f(x|W)\|/\alpha} \right)},$$

(4) 其中 α 表示学生-t 分布的自由度数。我们在本节后面讨论合适的 α 设置。

成本函数 C （使用上述 q 定义）的最小化可以通过反向传播实现，其中网络初始化时采用我们在 2.1 中描述的过程。微调所需的梯度由下式给出

$$\frac{\delta C}{\delta W} = \frac{\delta C}{\delta f(x|W)} \frac{\delta f(x|W)}{\delta W},$$

其中 δW 是通过标准反向传播计算的， δf 为

$$\frac{\delta C}{\delta f(x|W)} = \frac{2\alpha + 2}{\alpha} \sum_j (p_{ij} - q_{ij}) \left(\frac{f(x|W) - f(x|W)}{1 + \|f(x|W) - f(x|W)\|/\alpha} \right).$$

由于 p 和 q 的数量随批次中数据点数量成平方增长，成本函数的最小化通常必须使用几千个点的批次来实现（由于内存限制，通常无法使用更大的批次）。每次批次梯度计算后，解都会更新。

既然我们已经完全定义了参数化 t-SNE，接下来我们转向如何设置 α 的自由度数的问题。潜空间中使用的 Student-t 分布可能包含该分布下很大一部分概率质量，因为潜空间 Y 的体积随其维数呈指数增长。这导致了问题，可以通过将自由度 α 设置以纠正潜空间体积指数增长的方式来解决，因为增加自由度 α 会导致分布尾部变轻。事实上，参数 α 决定了潜在空间被“填满”的程度： α 值越低，数据中自然聚类之间的潜在空间间距越大，因为这些聚类产生了潜空间中不同数据点之间的更强排斥力。相比之下， α 值越高，数据中自然聚类之间的间隔越小，因此潜伏空间中有更多空间可用于适当建模数据的局部结构。下面，我们将讨论三种设置 Student-t 分布自由度的方法，该分布用于测量潜空间中的两两相似性。1) 固定金额。第一种方法是使用固定设置

of $\alpha = 1$, as is done by van der Maaten and Hinton (2008). This setting is likely to be subject to the problem with the heavy tails discussed above, as a fixed value of α does not correct for the exponential growth of the volume of the latent space (in higher dimensionalities).

2) *Linear relation.* As the thickness of the tail of a Student-t distribution decreases exponentially with the degrees of freedom α (and the volume of the latent space increases exponentially with the dimensionality of the latent space), it seems likely that the parameter setting for degrees of freedom α should be linearly dependent on the dimensionality d of the latent space. Hence, it seems reasonable to set $\alpha = d - 1$ in order to obtain a single degree of freedom in two-dimensional latent spaces, following van der Maaten and Hinton (2008)).

3) *Learning.* A possible problem of the second approach is that the appropriate value of α does not only depend on the dimensionality of the latent space. In fact, the most appropriate setting of α depends on the magnitude of the crowding problem, which in turn depends on the ratio between the *intrinsic dimensionality* of the data and the dimensionality of the latent space. For instance, if the intrinsic dimensionality is equal to the dimensionality of the latent space, the crowding problem does not occur at all, and the most appropriate value is thus $\alpha = \infty$ (note that a Student-t distribution with infinite degrees of freedom is equal to a Gaussian distribution). As the intrinsic dimensionality of the data at hand is usually unknown, the third approach treats α as a free parameter that should be optimized with respect to the cost function as well. The required gradient of the cost function C with respect to α is given by

$$\frac{\delta C}{\delta \alpha} = \sum_{i \neq j} \left(\frac{(-\alpha - 1)d_{ij}^2}{2\alpha^2 \left(1 + \frac{d_{ij}^2}{\alpha}\right)} + \frac{1}{2} \log \left(1 + \frac{d_{ij}^2}{\alpha}\right) \right) (p_{ij} - q_{ij}),$$

where d_{ij}^2 represents $\|f(x_i|W) - f(x_j|W)\|^2$. In the following section, we present experiments in which we used three different approaches for setting α .

3 EXPERIMENTS

In order to evaluate the performance of parametric t-SNE and to compare the three different settings for its parameter α , we performed experiments with parametric t-SNE on three datasets. The setup of these experiments is discussed in 3.1. The results of the experiments are presented in 3.2.

3.1 EXPERIMENTAL SETUP

We performed experiments on three datasets: (1) the MNIST dataset, (2) the characters dataset, and (3) the 20 newsgroups dataset. The MNIST dataset contains 70,000

images of handwritten digits of size 28×28 pixels. The dataset has a fixed division into 60,000 training images and held out 10,000 test images. The characters dataset consists of 40,121 grayscale images of handwritten upper-case characters and numerals of size 90×90 pixels, of which we used 35,000 images as training data and the remainder as test data. The characters dataset comprises 35 classes, viz., 10 numeric classes and 25 alpha classes (the character ‘X’ is missing in the dataset). The 20 newsgroups dataset contains 100-dimensional binary word-occurrence features for 16,242 documents gathered from 20 different newsgroups. We used 15,000 documents as training data, and the remaining documents as test data.

In our experiments, we compared parametric t-SNE with two other unsupervised parametric techniques for dimensionality reduction, viz., PCA and multilayer autoencoders (Hinton and Salakhutdinov, 2006). We also compared parametric t-SNE to NCA (Goldberger et al., 2005), which is a supervised linear dimensionality reduction technique. We evaluated the performance of the techniques by means of plotting two-dimensional visualizations, measuring generalization performances of nearest-neighbor classifiers, and evaluating the trustworthiness (Venna and Kaski, 2006) of the low-dimensional embeddings³. In order to make the comparison between parametric t-SNE and autoencoders as fair as possible, we used the same layout for both neural networks (where it should be noted that a parametric t-SNE network does not have the decoder part of an autoencoder). Motivated by the experimental setup employed by Salakhutdinov and Hinton (2007), we used $28 \times 28 - 500 - 500 - 2000 - d$ parametric t-SNE networks and autoencoders in our experiments on the MNIST dataset (where d represents the dimensionality of the latent space). In our experiments on the characters dataset, we used $90 \times 90 - 500 - 500 - 2000 - d$ networks. On the 20 newsgroups dataset, we used $100 - 150 - 150 - 500 - d$ networks. The autoencoders were trained using the same three-stage training approach as parametric t-SNE, but the autoencoder is finetuned by performing backpropagation as to minimize the sum of squared errors between the input and the output of the autoencoder (see (Hinton and Salakhutdinov, 2006) for details).

We used exactly the same procedure and parameter settings in the pretraining of the parametric t-SNE networks and the autoencoders. In the training of the RBMs whose hidden units have sigmoid activation functions (the RBMs in the first three layers), the learning rate is set to 0.1 and the

³The trustworthiness expresses to what extent the local structure of data is retained in a low-dimensional embedding in a value between 0 and 1. Mathematically, it is defined as $T(k) = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{j \in U_i^{(k)}} (r(i, j) - k)$, where $r(i, j)$ represents the rank of the low-dimensional datapoint j according to the pairwise distances between the low-dimensional datapoints, and $U_i^{(k)}$ represents the set of points that are among the k nearest neighbors in the low-dimensional space but not in the high-dimensional space (Venna and Kaski, 2006).

$\alpha = 1$ ，正如 van der Maaten 和 Hinton (2008) 所做的。该设置可能面临上述重尾的问题，因为固定值 α 无法校正潜空间体积的指数增长（在高维环境中）。

2) 线性关系。由于学生分布尾部厚度随自由度 α 指数级减小（且潜空间体积随潜在空间维数指数增长），自由度 α 参数设置很可能线性依赖于潜空间的维数 d 。因此，按照 van der Maaten 和 Hinton (2008) 的观点，设 $\alpha = d - 1$ 以获得二维潜空间中的单自由度似乎是合理的。

3) 学习。第二种方法的一个可能问题是， α 的适当值不仅取决于潜空间的维数。事实上，最合适的 α 设置取决于拥挤问题的大小，而拥挤问题又取决于数据的内在维度与潜空间维度之间的比值。例如，如果内在维度等于潜在空间的维数，则不会出现拥挤问题，因此最合适的值是 $\alpha = \infty$ （注意，具有无限自由度的学生-t 分布等于高斯分布）。由于手头数据的内在维度通常未知，第三种方法将 α 视为自由参数，应针对成本函数进行优化。成本函数 C 相对于 α 所需的梯度为

$$\frac{\partial C}{\partial \alpha} = \sum_{i \neq j} \frac{B_i B_j}{2\alpha} \frac{(-\alpha - 1)}{1 + \frac{d_{ij}^2}{\alpha}} \left(\frac{1}{2} \log \frac{1}{1 + \frac{d_{ij}^2}{\alpha}} + \frac{d_{ij}^2}{\alpha} \frac{1}{A} \right) (p - q),$$

其中 d 表示 $\|f(x|W) - f(x|W')\|$ 。在接下来的部分中，我们将展示使用三种不同方法来设置 α 的实验。

3 实验

为了评估参数化 t-SNE 的性能并比较其参数 α 的三种不同设置，我们在三个数据集上进行了参数化 t-SNE 实验。这些实验的设置将在 3.1 中进行了讨论。实验结果见 3.2。

3.1 实验装置

我们在三个数据集上进行了实验：(1) MNIST 数据集，(2) 角色数据集，(3) 20 个新闻组数据集。MNIST 数据集包含 70 个

000 张手写数字图片，大小为 28×28 像素。该数据集固定划分为 6 万张训练图像，并持有 1 万张测试图像。字符数据集包含 40,121 张手写大写字母和大小 90×90 像素数字的灰度图像，其中 35,000 张作为训练数据，其余作为测试数据。字符数据集包含 35 个类别，即 10 个数字类别和 25 个字母类别（数据集中缺少字符“X”）。20 个新闻组数据集包含了 16,242 份来自 20 个不同新闻组的文档的 100 维二进制词出现特征。我们使用了 15,000 份文档作为训练数据，其余文档作为测试数据。

在我们的实验中，我们将参数化 t-SNE 与另外两种无监督参数降维技术（即 PCA 和多层自编码器）进行比较（Hinton 和 Salakhutdinov, 2006）。我们还比较了参数化 t-SNE 与 NCA（Goldberger 等, 2005），后者是一种监督式线性降维技术。我们通过绘制二维可视化、测量最近邻分类器的泛化表现以及评估低维嵌入的可信度（Venna 和 Kaski, 2006）来评估技术的性能。为了尽可能公平地比较参数化 t-SNE 和自编码器，我们对两种神经网络采用了相同的布局（需要注意的是，参数化 t-SNE 网络没有自编码器的解码器部分）。受 Salakhutdinov 和 Hinton (2007) 实验设置的启发，我们在 MNIST 数据集上的实验中使用了 $28 \times 28 - 500 - 500 - 2000 - d$ 参数的 t-SNE 网络和自编码器（其中 d 代表潜空间的维数）。在我们对特征数据集的实验中，使用了 $90 \times 90 - 500 - 500 - 2000 - d$ 网络。在 20 个新闻组数据集中，我们使用了 $100 - 150 - 150 - 500 - d$ 网络。自编码器采用与参数 t-SNE 相同的三阶段训练方法进行训练，但通过进行反向传播进行微调，以最小化输入与输出之间的误差平方和（详见 Hinton 和 Salakhutdinov, 2006）。

我们在参数化 t-SNE 网络和自编码器的预训练中使用了完全相同的过程和参数设置。在训练隐藏单元具有 S 形激活函数的 RBMs（前三层的 RBM）时，学习率设为 0.1，且

可信度表示在低维嵌入中，数据的局部结构在 0 到 1 之间的值中被保留的程度。数学上定义为 $T(k) = 1 -$

$\frac{1}{n} \sum_{i=1}^n \left(\frac{r_{(i)}^2 - r_{(i+1)}^2}{r_{(i)}^2 - r_{(i+1)}^2} \right) \frac{1}{k}$ ，其中 $r_{(i)}$ 表示低维数据点 j 的秩，基于低维数据点之间的两对距离， U_i 表示在低维空间中处于 k 个最近邻中但不在高维空间中的点集合（Venna 和 Kaski, 2006）。

weight decay is set to 0.0002. The training of the RBMs with a linear activation function in the hidden units (the RBMs in the fourth layer) is performed using a learning rate of 0.01 and a weight decay of 0.0002. In the training of all RBMs, the momentum is set to 0.5 for the first five iterations, and to 0.9 afterwards. The RBMs are all trained using 50 iterations of contrastive divergence with one complete Gibbs sweep per iteration.

Both parametric t-SNE and the autoencoders were fine-tuned using 30 iterations of backpropagation using conjugate gradients on batches of 5,000 datapoints. The subdivision of training data into batches was fixed in order to facilitate the precomputation of the P matrices that are required in parametric t-SNE. In the experiments with parametric t-SNE, the variance σ_i of the Gaussian distributions was set such that the perplexity of the conditional distributions P_i was equal to 30. NCA was trained by running conjugate gradients on batches of 5,000 datapoints for 10 iterations using conjugate gradients.

3.2 RESULTS

In Figure 2, we present the visualizations of the MNIST dataset that were constructed by PCA, an autoencoder, and a parametric t-SNE network (using $\alpha = 1$). The visualizations were constructed by transforming the MNIST test images, that were held out during training, to two dimensions using the trained models. The results reveal the strong performance of parametric t-SNE compared to PCA and autoencoders. In particular, the PCA visualization mixes up most of the natural classes in the data. The autoencoder outperforms PCA, but cannot successfully separate the classes 4, 9, 6, and 8. In contrast, parametric t-SNE clearly separates all classes (although the visualization contains some debris that is mainly due to the presence of distorted digits in the data). In Table 1, we present the generalization errors of 1-nearest neighbor classifiers that were trained on the low-dimensional representations obtained from the three parametric dimensionality reduction techniques (using three different dimensionalities for the latent space). The generalization errors were measured on test data that was held out during the training of both the dimensionality reduction techniques and the classifiers. The corresponding trustworthinesses $T(12)$ of the embeddings are presented in Table 2. In both tables, the best performance in each experiment is typeset in boldface. From the results presented in Table 1 and 2, we can make the following two observations.

First, we observe that parametric t-SNE performs better or on par with the other techniques in all experiments. In particular, the performance of parametric t-SNE is very strong if the dimensionality of the latent space is not large enough to accommodate for all properties of the data. In this case, the heavy tails of the distribution of parametric t-SNE in the latent space push the natural clusters in the data apart,

whereas PCA and autoencoders construct embeddings in which these natural clusters (partially) overlap. The high trustworthinesses of the parametric t-SNE embeddings indicate that parametric t-SNE preserves the local structure of the data in the latent space well. The results reveal that parametric t-SNE also outperforms linear NCA, even though NCA has the advantage of being fully supervised. Second, we observe that it is disadvantageous to use a single degree of freedom in the latent space if that latent space has more than, say, two dimensions. Our results reveal that it is better to use make the number of degrees of freedom α linearly dependent on the dimensionality of the latent space d , for reasons we already explained in 2.2. The results also show that learning the appropriate number of degrees of freedom α leads to similar results. The learned value of α was usually slightly smaller than $d - 1$ in our experiments.

4 DISCUSSION

From the results of our experiments, we observe that parametric t-SNE often outperforms two other unsupervised parametric dimensionality reduction techniques, in particular, if the dimensionality of the latent space is relatively low. These results are due to the main differences of parametric t-SNE compared to PCA and autoencoders, which we discuss below.

The strong performance of parametric t-SNE compared to PCA can be explained from the two main problems of PCA. First, the linear nature of PCA is too restrictive for the technique to find appropriate embeddings for non-linear real-world data. Second, PCA focuses primarily on retaining large pairwise distances in the latent space (which can be understood from its relation to classical scaling), whereas it is more important to retain the local structure of the data in the latent space.

The strong performance of parametric t-SNE compared to autoencoders, especially if the latent space has a relatively low dimensionality, can be understood from the following difference between parametric t-SNE and autoencoders. Parametric t-SNE aims to model the local structure of the data appropriately in the latent space, and it attempts to create separation between the natural clusters in the data (by means of the heavy-tailed distribution in the latent space). In contrast, autoencoders mainly aim to maximize the variance of the data in the latent space, in order to achieve low reconstruction errors. As a result of the maximization of the variance, autoencoders generally do not construct low-dimensional data representations in which the natural classes in the data are widely separated (as this would decrease the variance of the low-dimensional data representation, and increase the reconstruction error). The relatively poor separation between natural classes in low-dimensional data representations constructed by autoencoders leads to inferior generalization performance of nearest neighbors classifiers compared to parametric t-SNE, in

权重衰减设置为 0.0002。在隐藏单元（第四层的 RBM）中，用线性激活函数训练 RBMs，学习率为 0.01，权重衰减为 0.0002。在所有 RBM 的训练中，前五次迭代的动量设置为 0.5，之后则为 0.9。所有 RBM 均通过 50 次对比发散迭代训练，每次迭代完成一次完整的吉布斯扫描。

参数化 t-SNE 和自编码器均通过 30 次反向传播和共轭梯度对 5,000 个数据点的批次进行微调。训练数据被固定为批次，以便于参数化 t-SNE 所需的 P 矩阵的预计算。在参数 t-SNE 实验中，高斯分布的方差 σ 被设定为条件分布 P 的困惑度为 30。NCA 通过对 5,000 个数据点批量运行共轭梯度，进行 10 次迭代，使用共轭梯度进行训练。

3.2 结果

图 2 展示了由 PCA、自编码器和参数化 t-SNE 网络（使用 $\alpha=1$ ）构建的 MNIST 数据集可视化。这些可视化是通过将训练中展示的 MNIST 测试图像转换为二维，使用训练好的模型构建的。结果显示参数 t-SNE 相较于 PCA 和自编码器表现出优异的性能。特别是，PCA 可视化混合了数据中的大多数自然类。自编码器性能优于 PCA，但无法成功分离 4、9、6 和 8 类。相比之下，参数化 t-SNE 清晰地区分所有类别（尽管可视化中存在一些杂质，主要因数据中数字失真所致）。在表 1 中，我们展示了基于三种参数降维技术（使用三种不同维度的潜在空间）获得的低维表示训练的 1 最近邻分类器的推广误差。泛化误差是在降维技术和分类器训练过程中展示的测试数据上测量的。嵌入的对应可信度 T (12) 见表 2。在这两个表格中，每个实验中最佳表现都以加粗字体排版。根据表 1 和表 2 中呈现的结果，我们可以做出以下两点观察。

首先，我们观察到参数化 t-SNE 在所有实验中表现更好或与其他技术相当。特别地，如果潜在空间的维数不足以涵盖数据的所有性质，参数化 t-SNE 的性能将非常强。在这种情况下，潜在空间中参数 t-SNE 分布的重尾将数据中的自然聚类推开，

而 PCA 和自编码器则构建这些自然聚类（部分）重叠的嵌入。参数化 t-SNE 嵌入的高可信度表明参数化 t-SNE 很好地保留了潜空间中数据的局部结构。结果显示，参数化 t-SNE 也优于线性 NCA，尽管 NCA 具有完全监督的优势。其次，我们观察到，如果潜空间的维度超过比如说，使用单一自由度是不利的。我们的结果表明，最好将自由度 α 线性地依赖于潜空间 d 的维数，原因我们在 2.2 中已解释过。结果还表明，学习适当的自由度 α 能产生类似的结果。在我们的实验中， α 的学习值通常略小于 d-1。

4 讨论

根据实验结果，我们观察到参数化 t-SNE 通常优于另外两种无监督参数降维技术，尤其是在潜空间维度相对较低的情况下。这些结果源于参数 t-SNE 与 PCA 和自编码器的主要差异，我们将在下文讨论。

参数 t-SNE 相较于 PCA 表现优异，可归因于 PCA 的两个主要问题。首先，PCA 的线性特性过于限制，无法为非线性现实世界数据找到合适的嵌入。其次，主成分分析主要关注于保留潜空间中的较大两两距离（可从其与经典缩放的关系中理解），而更重要的是保留潜空间中数据的局部结构。

参数化 t-SNE 相较于自编码器的性能，尤其是在潜在空间维度较低的情况下，可以通过参数化 t-SNE 和自编码器的以下区别来理解。参数化 t-SNE 旨在适当地模拟数据在潜在空间中的局部结构，并试图通过潜空间中的重尾分布在数据中创建自然聚类之间的隔离。相比之下，自编码器主要旨在最大化潜空间中数据的方差，以实现较低的重建误差。由于方差最大化，自编码器通常不会构建数据中自然类别广泛分离的低维数据表示（因为这会降低低维数据表示的方差，增加重建误差）。由自编码器构建的低维数据表示中自然类之间的相对较差的分离，导致最近邻分类器的推广性能不如参数化的 t-SNE，在

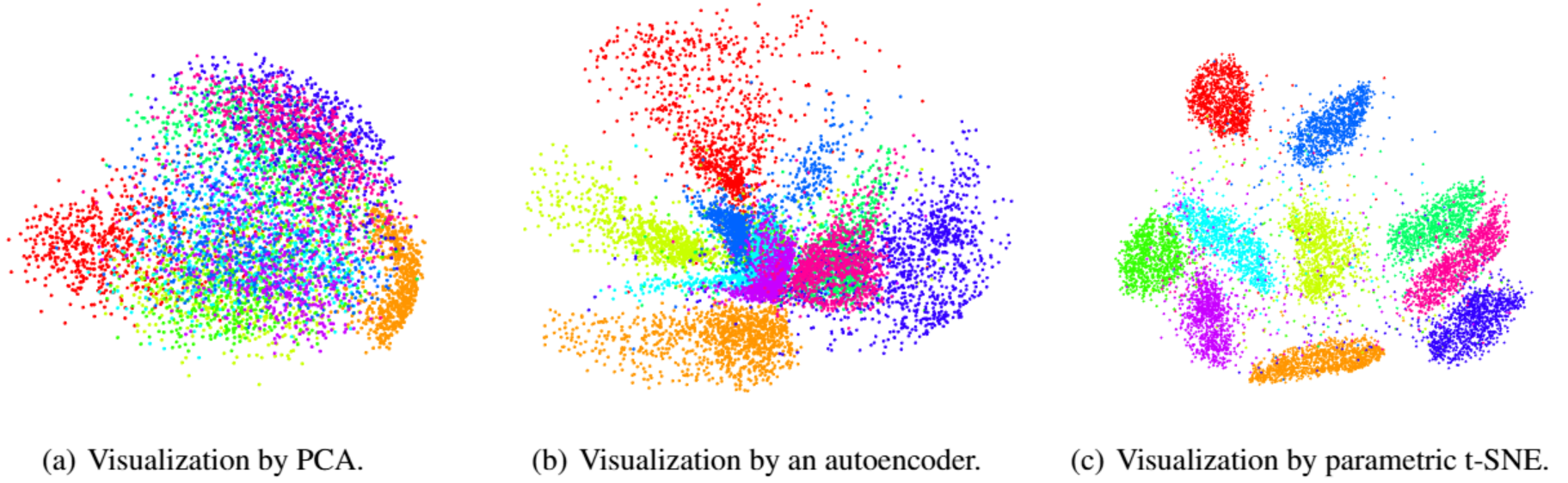


Figure 2: Visualizations of 10,000 digits from the MNIST dataset by parametric dimensionality reduction techniques.

	MNIST			Characters			20 Newsgroups		
	2D	10D	30D	2D	10D	30D	2D	10D	30D
PCA	78.16%	43.03%	10.78%	86.72%	60.73%	20.50%	35.99%	27.05%	28.82%
NCA	56.84%	8.84%	7.32%	72.90%	24.68%	17.95%	30.76%	26.65%	26.09%
Autoencoder	66.84%	6.33%	2.70%	82.93%	17.91%	11.11%	37.60%	29.15%	27.62%
Par. t-SNE, $\alpha = 1$	9.90%	5.38%	5.41%	43.90%	26.01%	23.98%	34.30%	24.40%	24.88%
Par. t-SNE, $\alpha = d - 1$	9.90%	4.58%	2.76%	43.90%	17.13%	13.55%	35.10%	25.28%	23.75%
Par. t-SNE, learned α	12.68%	4.85%	2.70%	44.78%	17.30%	14.31%	33.82%	27.21%	24.72%

Table 1: Generalization errors of 1-nearest neighbor classifiers on low-dimensional representations of the MNIST dataset, the characters dataset, and the 20 newsgroups dataset.

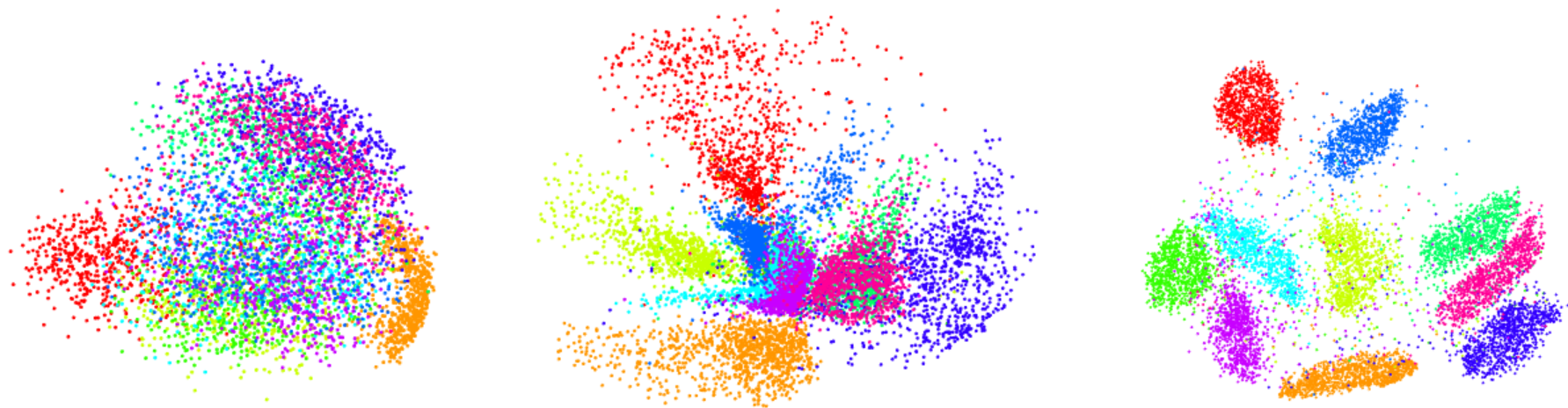
particular, if the dimensionality of the latent space is relatively low. Moreover, parametric t-SNE provides computational advantages over autoencoders. An autoencoder consists of an encoder part and a decoder part, whereas parametric t-SNE only employs an encoder network. As a result, errors have to be backpropagated through half the number of layers in parametric t-SNE (compared to autoencoders), which gives it an computational advantage over autoencoders (even though the computation of the errors is somewhat more expensive in parametric t-SNE).

A notable advantage of autoencoders is that they provide the capability to reconstruct the original data from its low-dimensional representation in the latent space. In other words, autoencoders do not only provide a parametric mapping from the data space to the latent space, but also the other way around. A possible approach to address this shortcoming is to use the decoder part of an autoencoder as a regularizer on the parametric t-SNE network, i.e., to minimize a weighted sum of Equation 3 and the reconstruction error (as is done for non-linear NCA by Salakhutdinov and Hinton (2007)).

As the number of parameters in parametric t-SNE and autoencoders is larger than in PCA, these techniques are likely to be more susceptible to overfitting. However, we did not observe overfitting effects in our experiments, probably because of the relatively large number of instances in our training data. If parametric t-SNE or autoencoders are

trained on smaller datasets, it may be necessary to use early stopping (Caruana et al., 2001).

The results of our experiments not only reveal the strong performance of parametric t-SNE compared to PCA and autoencoders, but also provide insight into the nature of the crowding problem. In particular, the results reveal that the severity of the crowding problem depends on the ratio between the intrinsic dimensionality of the data and the dimensionality of the latent space. The number of degrees of freedom α should thus be set accordingly. We suggested to treat α as a parameter that has to be learned as well, and although competitive, learning α does not always outperform a setting in which α depends linearly on the dimensionality of the latent space. Presumably, this observation is due to the following. When α is learned, it is set in such a way as to ‘fill up’ the latent space. This decreases the Kullback-Leibler divergence that parametric t-SNE minimizes, because it provides more space to model the local structure of the data appropriately (recall that the cost function focuses on retaining local structure). Although the ‘filling up’ of the space is advantageous for modeling the local structure of the data (as is illustrated by the high trustworthinesses when α is learned), it has a negative influence on the generalization performance of nearest neighbor classifiers on the low-dimensional data representation, as it decreases the separation between the natural clusters in the data.



(a) PCA 的可视化。(b) 通过自编码器进行可视化。(c) 通过参数化 t-SNE 进行可视化。

图 2：通过参数降维技术对 MNIST 数据集中 10,000 位数字的可视化。

	MNIST 角色 20 个新闻组											
	2D	10D	30D	2D	10D	30D	2D	10D	30D			
PCA	78.16%	43.03%	10.78%	86.72%	60.73%	20.50%	35.99%	27.05%	28.05%	82%	NCA	56.84%
	24.68%	17.95%	30.76%	26.65%	26.09%	自动编码器	66.84%	6.33%	2.70%	82.93%		8.84%
	27.62%	部分 t-SNE, $\alpha = 1$	9.90%	5.38%	5.41%	43.90%	26.01%	23.98%	34.30%	24.40%	24.88%	7.32%
	4.58%	2.76%	43.90%	17.13%	13.55%	35.10%	25.28%	23.75%	部分 t-SNE, 勒恩·A	12.68%	4.85%	72.90%
	14.31%	33.82%	27.21%	24.72%								29.15%

表 1：MNIST 数据集、角色数据集和 20 新闻组数据集的低维表示中，1 最近邻分类器的推广误差。

特别的，如果潜空间的维数相对较低。此外，参数化 t-SNE 在计算上优于自编码器。自编码器由编码器部分和解码器部分组成，而参数化 t-SNE 仅使用编码器网络。因此，参数化 t-SNE 中错误的反向传播层数是自编码器的一半，这使其在计算上优于自编码器（尽管参数 t-SNE 的错误计算成本略高）。

自编码器的一个显著优势是能够从潜空间中的低维表示重建原始数据。换句话说，自编码器不仅提供数据空间到潜在空间的参数映射，反过来也提供映射。解决这一缺陷的一种可能方法是将自编码器的译码器部分作为参数 t-SNE 网络上的正则化器，即最小化方程 3 与重建误差的加权和（如 Salakhutdinov 和 Hinton（2007）对非线性 NCA 所做的那样）。

由于参数化 t-SNE 和自编码器的参数数量比 PCA 多，这些技术更易发生过拟合。然而，我们的实验中并未观察到过拟合效应，可能是因为训练数据中出现的实例数量相对较多

如果参数化 t-SNE 或自编码器是在较小数据集上训练的，可能需要使用早期停止（Caruana 等，2001）。我们的实验结果不仅揭示了参数 t-SNE 相较于 PCA 和自编码器的优异表现，也为拥挤问题的本质提供了见解。特别地，结果表明拥挤问题的严重程度取决于数据的内在维度与潜空间维度之间的比值。因此， α 的自由度数应相应设定。我们建议将 α 也视为必须学习的参数，尽管具有竞争力，学习 α 并不总是优于 α 线性依赖于潜空间维度的环境。这一观察大概源于以下原因。当 α 被学习时，它会被设定成“填满”潜在空间的方式。这减少了参数 t-SNE 最小化的 KullbackLeibler 散度，因为它提供了更多空间来适当建模数据的局部结构（回想成本函数侧重于保留局部结构）。虽然“填充”空间有助于建模数据的局部结构（如学习 α 时的高可信度所示），但由于减少了数据中自然聚类之间的分离，它对最近邻分类器在低维数据表示上的推广性能有负面影响。

	MNIST			Characters			20 Newsgroups		
	2D	10D	30D	2D	10D	30D	2D	10D	30D
PCA	0.744	0.991	0.998	0.735	0.971	0.994	0.634	0.847	0.953
NCA	0.721	0.968	0.971	0.721	0.935	0.957	0.633	0.705	0.728
Autoencoder	0.729	0.996	0.999	0.721	0.976	0.992	0.612	0.856	0.961
Par. t-SNE, $\alpha = 1$	0.926	0.983	0.983	0.866	0.957	0.959	0.720	0.854	0.866
Par. t-SNE, $\alpha = d - 1$	0.927	0.997	0.999	0.866	0.988	0.995	0.714	0.864	0.942
Par. t-SNE, learned α	0.921	0.996	0.999	0.861	0.988	0.995	0.722	0.857	0.941

Table 2: Trustworthiness $T(12)$ of low-dimensional representations of the MNIST dataset, the characters dataset, and the 20 newsgroups dataset.

5 CONCLUSIONS

We have shown how a deep feed-forward neural network can be trained that reduces the dimensionality of data, while preserving its local structure. The results of our experiments with parametric t-SNE on three datasets showed that it outperforms other unsupervised parametric dimensionality reduction techniques such as autoencoders. A Matlab implementation of parametric t-SNE is available from <http://ticc.uvt.nl/~lvdrmaaten/tsne>. In future work, we aim to investigate parametric t-SNE networks that use the decoder part of an autoencoder as a regularizer. Also, we aim to investigate how parametric t-SNE can be combined with supervised dimensionality reduction techniques to obtain better generalization performances in semi-supervised learning settings.

Acknowledgements

The author thanks Geoffrey Hinton for many helpful discussions, and Eric Postma for his comments. Laurens van der Maaten is supported by the Netherlands Organization for Scientific Research, project RICH (grant 640.002.401).

References

- Y. Bengio, J.-F. Paiement, P. Vincent, O. Delalleau, N. Le Roux, and M. Ouimet. Out-of-sample extensions for LLE, Isomap, MDS, eigenmaps, and spectral clustering. In *Advances in Neural Information Processing Systems*, volume 16, pages 177–184, 2004.
- R. Caruana, S. Lawrence, and L. Giles. Overtting in neural nets: Backpropagation, conjugate gradient, and early stopping. In *Advances of Neural Information Processing Systems*, volume 13, pages 402–408, 2001.
- A. Globerson, G. Chechik, F. Pereira, and N. Tishby. Euclidean embedding of co-occurrence data. *Journal of Machine Learning Research*, 8:2265–2295, 2007.
- J. Goldberger, S. Roweis, G.E. Hinton, and R.R. Salakhutdinov. Neighbourhood components analysis. In *Advances in Neural Information Processing Systems*, volume 17, pages 513–520, 2005.
- G.E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002.
- G.E. Hinton and S.T. Roweis. Stochastic Neighbor Embedding. In *Advances in Neural Information Processing Systems*, volume 15, pages 833–840, 2002.
- G.E. Hinton and R.R. Salakhutdinov. Reducing the dimensionality of data with neural networks. *Science*, 313(5786):504–507, 2006.
- T. Iwata, K. Saito, N. Ueda, S. Stromsten, T.L. Griffiths, and J.B. Tenenbaum. Parametric embedding for class visualization. *Neural Computation*, 19(9):2536–2556, 2007.
- H. Larochelle, Y. Bengio, J. Louradour, and P. Lamblin. Exploring strategies for training deep neural networks. *Journal of Machine Learning Research*, 10(Jan):1–40, 2009.
- N.D. Lawrence and J. Quiñero Candela. Local distance preservation in the GP-LVM through back constraints. In *Proceedings of the International Conference on Machine Learning*, pages 513–520, 2006.
- D. Lowe and M. Tipping. Feed-forward neural networks and topographic mappings for exploratory data analysis. *Neural Computing and Applications*, 4(2):83–95, 1996.
- R. Min. A non-linear dimensionality reduction method for improving nearest neighbour classification. Master’s thesis, University of Toronto, Canada, 2005.
- S.T. Roweis and L.K. Saul. Nonlinear dimensionality reduction by Locally Linear Embedding. *Science*, 290(5500):2323–2326, 2000.
- R.R. Salakhutdinov and G.E. Hinton. Learning a non-linear embedding by preserving class neighbourhood structure. In *Proceedings of the 11th International Conference on Artificial Intelligence and Statistics*, pages 412–419, 2007.
- J.B. Tenenbaum, V. de Silva, and J.C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000.
- L.J.P. van der Maaten and G.E. Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(Nov):2579–2605, 2008.
- J. Venna and S. Kaski. Visualizing gene interaction graphs with local multidimensional scaling. In *Proceedings of the 14th European Symposium on Artificial Neural Networks*, pages 557–562, 2006.
- K.Q. Weinberger, F. Sha, and L.K. Saul. Learning a kernel matrix for nonlinear dimensionality reduction. In *Proceedings of the 21st International Conference on Machine Learning*, pages 839–846, 2004.
- M. Welling, M. Rosen-Zvi, and G. Hinton. Exponential family harmoniums with an application to information retrieval. In *Advances in Neural Information Processing Systems*, volume 17, pages 1481–1488, 2004.

	MNIST 角色 20 个新闻组																		
	2D	10D	30D	2D	10D	30D	2D	10D	30D										
PCA	0.744	0.991	0.998	0.735	0.971	0.994	0.634	0.847	0.953	NCA	0.721	0.968	0.971	0.721	0.935	0.957	0.633	0.705	0.728
自编码器	0.729	0.996	0.999	0.721	0.976	0.992	0.612	0.856	0.961	参数。t-SNE, $4\alpha = 1$	0.926	0.983	0.983	0.866	0.957				
	0.959	0.720	0.854	0.866	Part-SNE, $4\alpha = d - 1$	0.927	0.997	0.999	0.866	0.988	0.995	0.714	0.864	0.942	Part-SNE, 学习 α				
	0.921	0.996	0.999	0.861		0.988	0.995	0.722	0.857	0.941									

表 2：MNIST 数据集、角色数据集和 20 个新闻组数据集低维表示的可信度 T（12）。

5 结论

我们展示了如何训练深度前馈神经网络，降低数据的维度，同时保持其局部结构。我们在三个数据集上使用参数化 t-SNE 实验结果显示，其性能优于其他无监督参数降维技术，如自编码器。<http://ticc.uvt.nl/lvdrmaaten/tsne> 提供了参数化 t-SNE 的 Matlab 实现。未来研究中，我们旨在研究使用自编码器解码器部分作为正则化器的参数化 t-SNE 网络。此外，我们还旨在探讨如何将参数化 t-SNE 与监督降维技术结合，以在半监督学习环境中获得更优的泛化表现。

致谢

作者感谢杰弗里·辛顿的许多有益讨论，以及埃里克·波斯特马的评论。Laurens van der Maaten 得到了荷兰科学组织 RICH 项目（资助号 640.002.401）的支持。

引用

Y. 本吉奥, J.-F.佩芒, P. 文森特, O. 德拉洛, N. 勒鲁, 和 M. 乌伊梅。LLE、同位图、MDS、特征图和谐聚类的样本外扩展。在《新进军》中
ral 信息处理系统, 第 16 卷, 第 177–184 页, 2004 年。

R. 卡鲁阿纳、S. 劳伦斯和 L. 贾尔斯。神经网络中的过载：反向传播、共轭梯度和早期停止。
载于《神经信息处理系统进展》，第 13 卷, 第 402–408 页, 2001 年。

A. 格洛伯森、G. 切奇克、F. 佩雷拉和 N. 蒂什比。共现数据的欧几里得嵌入。《机器学习研究杂志》，8: 2265–2295,2007 年。

J. 戈德伯格、S. 罗韦斯、G.E. 辛顿和 R.R. 萨拉胡特迪诺夫。
邻域组成分析。载于《神经信息处理系统进展》，第 17 卷, 第 513–520 页, 2005 年。

G.E. 欣顿。通过最小化对比度来训练专家的产品。《神经计算》，14 (8): 1771–1800, 2002.

G.E. 辛顿和 S.T. 罗伊斯。随机邻居嵌入。
载于《神经信息处理系统进展》，第 15 卷, 第 833–840 页, 2002 年。

G.E. 辛顿和 R.R. 萨拉胡丁诺夫。通过神经网络降低数据的维度。《科学》，313 (5786): 504–507,2006 年。岩田正、斋藤一、上田诚、斯特罗姆斯滕、格里菲斯和 J.B. 特姆。用于类可视化的参数嵌入。
《神经计算》，19 (9): 2536–2556,2007 年。

H. 拉罗谢尔、Y. 本吉奥、J. 卢拉杜尔和 P. 兰布林。探索深度神经网络训练策略。《机器学习研究杂志》，10 (1 月): 1–40,2009 年。

N.D. 劳伦斯和 J. 奎诺内罗·坎德拉。通过反约束实现 GP-LVM 的局部距离保持。在《继续》中——
国际机器学习会议, 第 513–520 页, 2006 年。

D. 洛威和 M. 蒂平。前馈神经网络和拓扑映射用于探索性数据分析。《神经计算与应用》，4 (2): 83–95,1996 年。R. Min。一种用于改进最近邻分类的非线性降维方法。硕士论文, 加拿大多伦多大学, 2005 年。S.T.罗伊斯和 L.K.索尔。通过局部线性嵌入实现非线性降维。《科学》，290 (5500): 2323–2326,2000 年。R.R. 萨拉库蒂诺夫和 G.E. 欣顿。通过保持类邻域结构学习非线性嵌入。职业版——
《第 11 届人工智能与统计国际会议的退出》，第 412–419 页, 2007 年。

J.B. 特姆、V. 德席尔瓦和 J.C. 兰福德。非线性降维的全局几何框架。《科学》，290 (5500): 2319–2323,2000 年。L.J.P. 范德马滕和 G.E. 欣顿。使用数据可视化
不。《机器学习研究杂志》，9 (11 月): 2579–2605,2008 年。

J. 文纳和 S. 卡斯基。通过局部多维尺度可视化基因相互作用图。在 14EU 会议纪要中——
Ropean 人工神经网络研讨会, 第 557–562 页, 2006 年。

K.Q. 温伯格、F. 沙和 L.K. 索尔。学习一个用于非线性降维的核矩阵。在
2004 年, 第 21 届国际机器学习会议, 第 839–846 页。

M. 韦林、M. 罗森-兹维和 G. 欣顿。指数族风琴及其在信息检索中的应用。
载于《神经信息处理系统进展》，第 17 卷, 第 1481–1488 页, 2004 年。